Laboratorium
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Laporan Akhir
Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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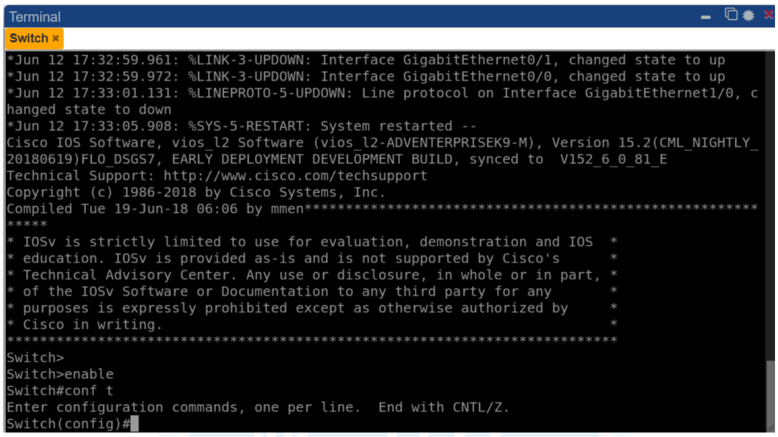
2026

1 Langkah Percobaan

Pada praktikum ini dilakukan konfigurasi routing dinamis OSPF pada jaringan multivendor yang terdiri dari Cisco Router, Huawei Router, dan MikroTik Router. Konfigurasi dilakukan untuk membangun pertukaran informasi routing secara otomatis serta menyediakan jalur redundan antarjaringan. Setelah konfigurasi selesai, dilakukan pengujian konektivitas dan failover dengan mensimulasikan gangguan pada link utama untuk memastikan OSPF dapat melakukan konvergensi dan mengalihkan trafik ke jalur cadangan secara otomatis.

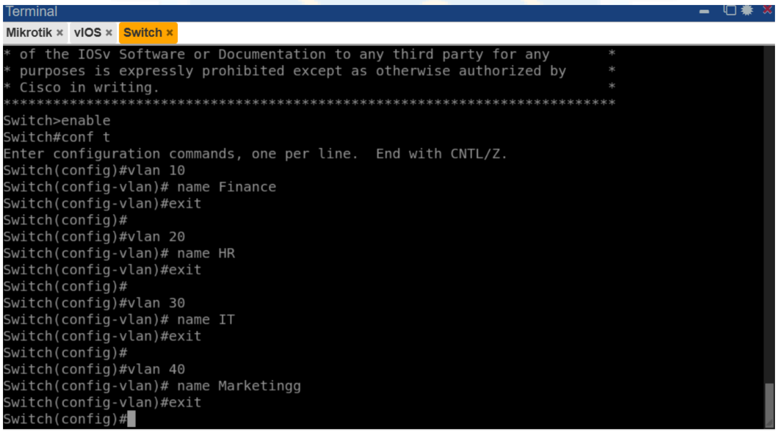
1.1 Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch



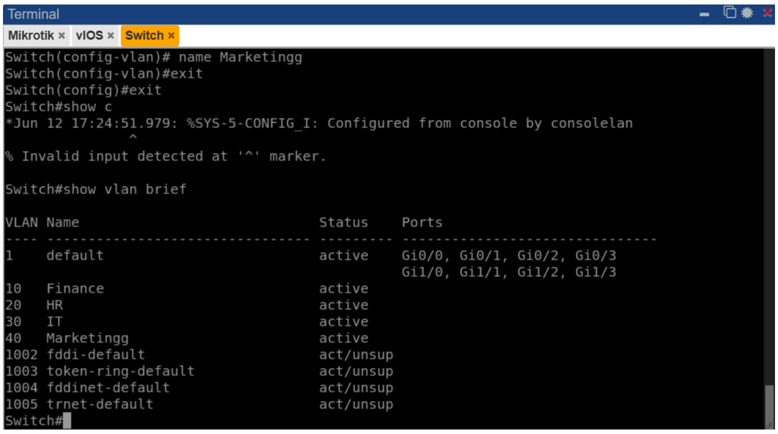
Gambar 1: Login Ke terimnal Cisco

2. Membuat Vlan



Gambar 2: Konfigurasi vlan cisco

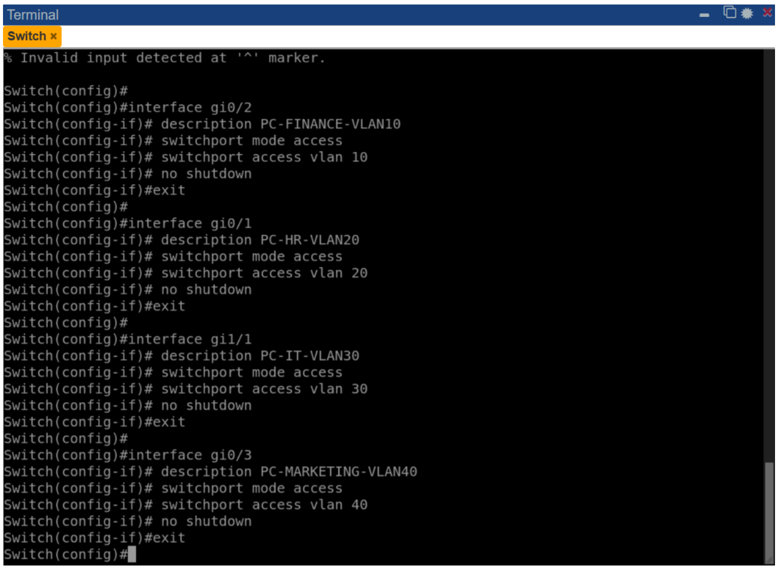
3. Pastikan vlan sudah dibuat dan statusnya aktif



```
Terminal
Mikrotik × vIOS × Switch ×
Switch(config-vlan)# name Marketingg
Switch(config-vlan)#exit
Switch(config)#exit
Switch#show c
*Jun 12 17:24:51.979: %SYS-5-CONFIG_I: Configured from console by consolelan
                ^
% Invalid input detected at '^' marker.

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
                                                Gi1/0, Gi1/1, Gi1/2, Gi1/3
10   Finance                          active
20   HR                               active
30   IT                               active
40   Marketingg                       active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch#
```

Gambar 3: vlan status aktif

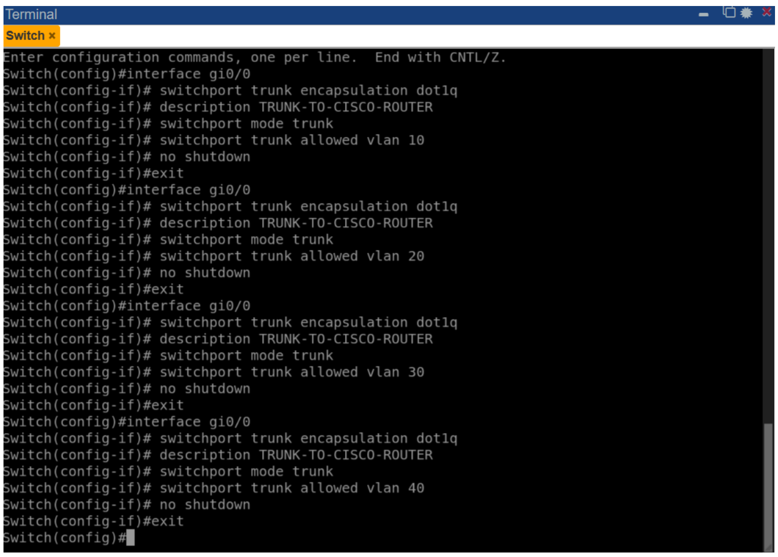
4. Konfigurasi Access port untuk setiap vlan



```
Terminal
Switch ×
% Invalid input detected at '^' marker.

Switch(config)#
Switch(config)#interface gi0/2
Switch(config-if)# description PC-FINANCE-VLAN10
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/1
Switch(config-if)# description PC-HR-VLAN20
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi1/1
Switch(config-if)# description PC-IT-VLAN30
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/3
Switch(config-if)# description PC-MARKETING-VLAN40
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
```

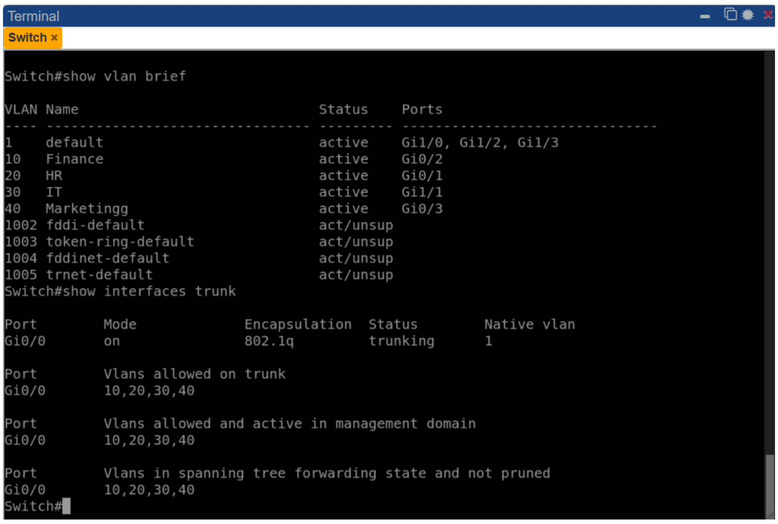
Gambar 4: Konfigurasi Access port

5. Konfigurasi Trunk ke Cisco Router



```
Terminal
Switch ×
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-TO-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-TO-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-TO-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-TO-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
```

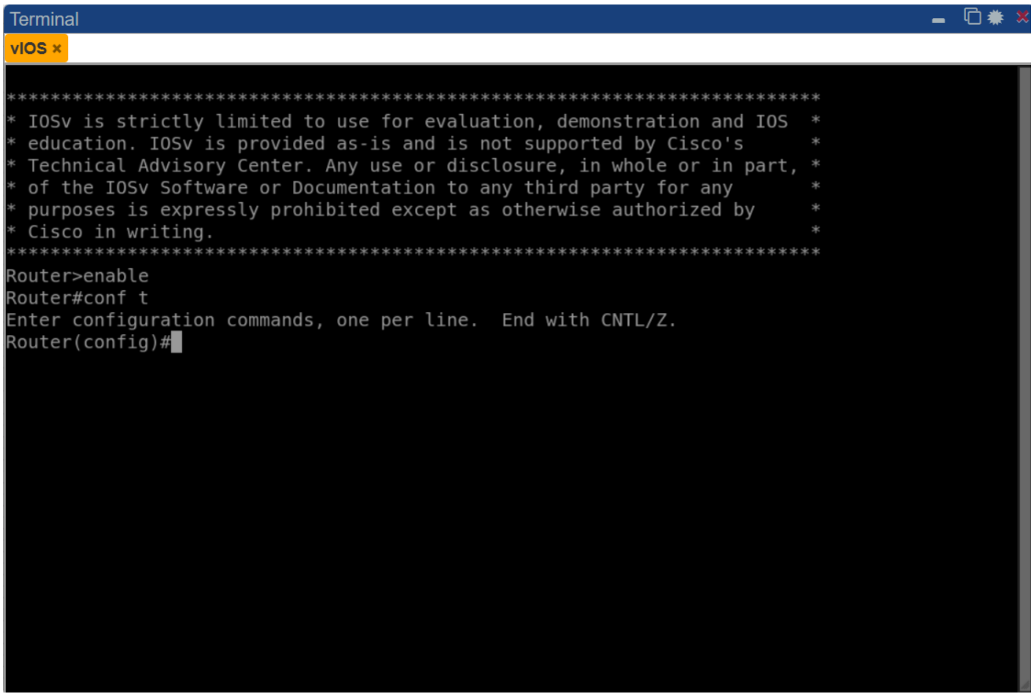
Gambar 5: Konfigurasi Trunk di router



Gambar 6: Hasil Show Trunk

1.2 Konfigurasi Cisco Router

1. Masuk ke Cisco Router



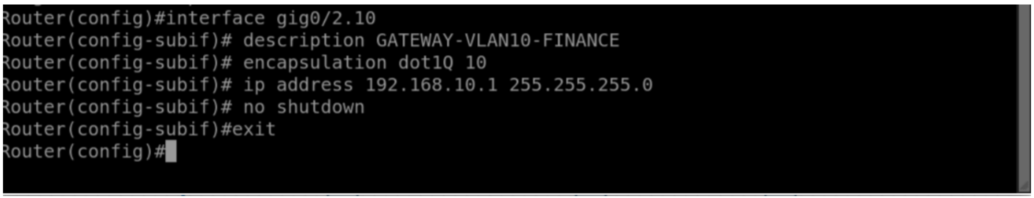
Gambar 7: Login ke cisco router

2. Konfigurasi Inteface Trunk ke Switch

```
Terminal
vIOS ×
**********************************************************************
* IOSv is strictly limited to use for evaluation, demonstration and IOS  *
* education. IOSv is provided as-is and is not supported by Cisco's      *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any       *
* purposes is expressly prohibited except as otherwise authorized by     *
* Cisco in writing.                                                      *
**********************************************************************
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)# description TRUNK-TO-CISCO-SWITCH
Router(config-if)# no ip address
Router(config-if)# no shutdown
Router(config-if)# exit
```

Gambar 8: Konfigurasi Trunk di switch

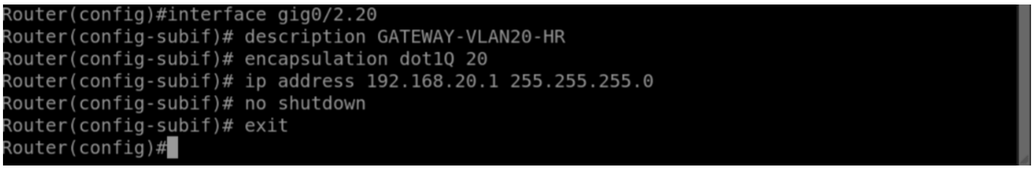
3. Konfigurasi Subinterface VLAN 10



```
Router(config)#interface gig0/2.10
Router(config-subif)# description GATEWAY-VLAN10-FINANCE
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)#exit
Router(config)#
```

Gambar 9: Konfigurasi Subinterface

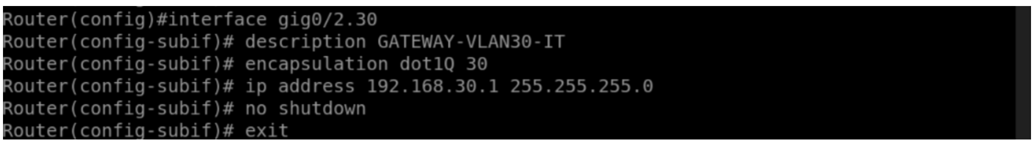
4. Konfigurasi Subinterface VLAN 20



```
Router(config)#interface gig0/2.20
Router(config-subif)# description GATEWAY-VLAN20-HR
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
Router(config)#
```

Gambar 10: Konfigurasi Subinterface

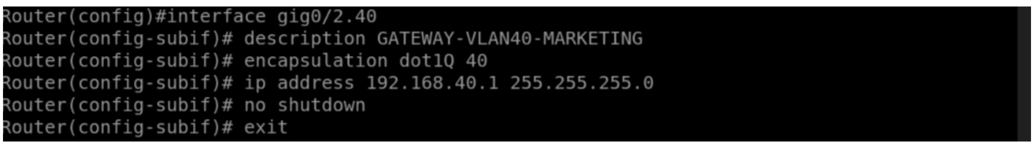
5. Konfigurasi Subinterface VLAN 30



```
Router(config)#interface gig0/2.30
Router(config-subif)# description GATEWAY-VLAN30-IT
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
```

Gambar 11: Konfigurasi Subinterface

6. Konfigurasi Subinterface VLAN 40



```
Router(config)#interface gig0/2.40
Router(config-subif)# description GATEWAY-VLAN40-MARKETING
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config-subif)# exit
```

Gambar 12: Konfigurasi Subinterface

7. Konfigurasi DHCP-Server untuk Vlan 10

```
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
```

Gambar 13: Konfigurasi DHCP-Server

8. Konfigurasi DHCP-Server untuk Vlan 20

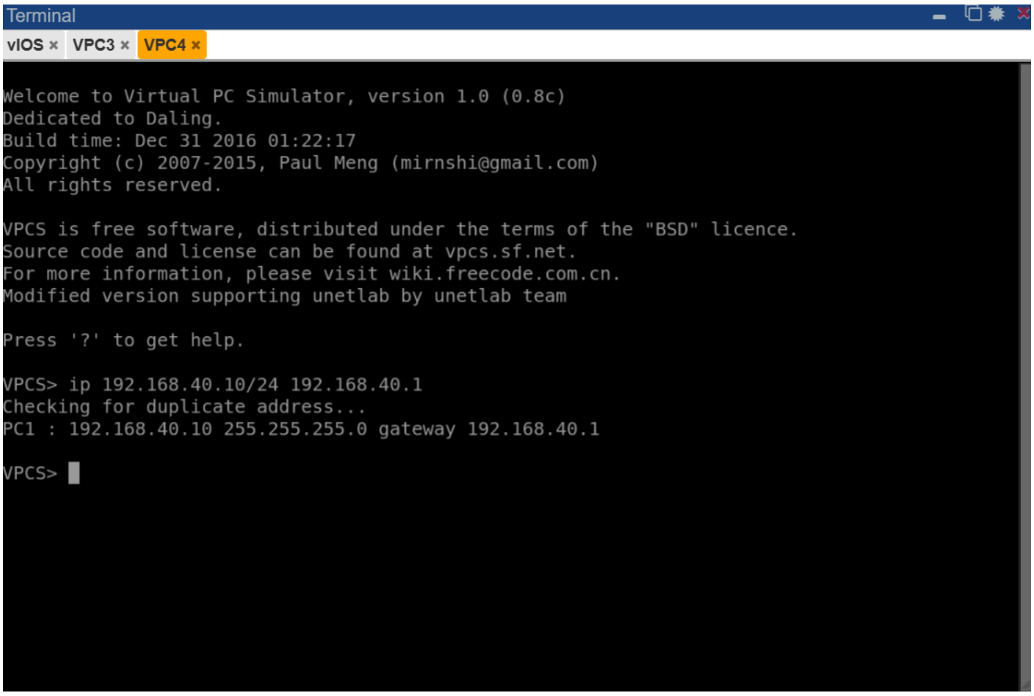
```
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
```

Gambar 14: Konfigurasi DHCP-Server

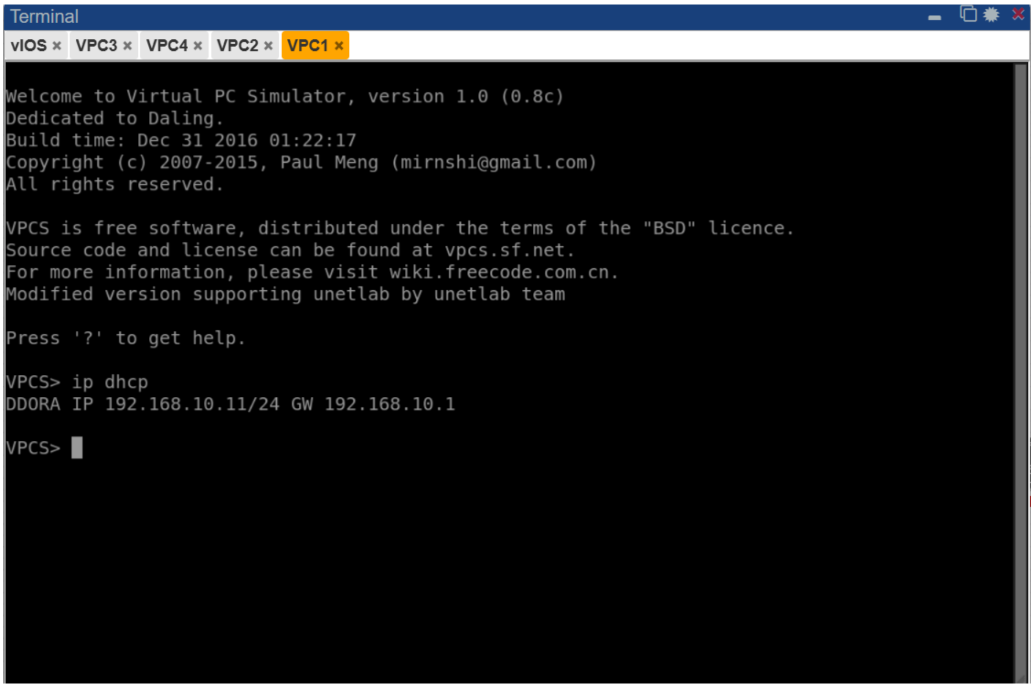
9. Konfigurasi IP Static pada VPCS untuk vlan 30 dan 40

```
Terminal
vIOS × VPC3 × VPC4 ×

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS>
```

Gambar 15: Konfigurasi IP Static pada vpcs3

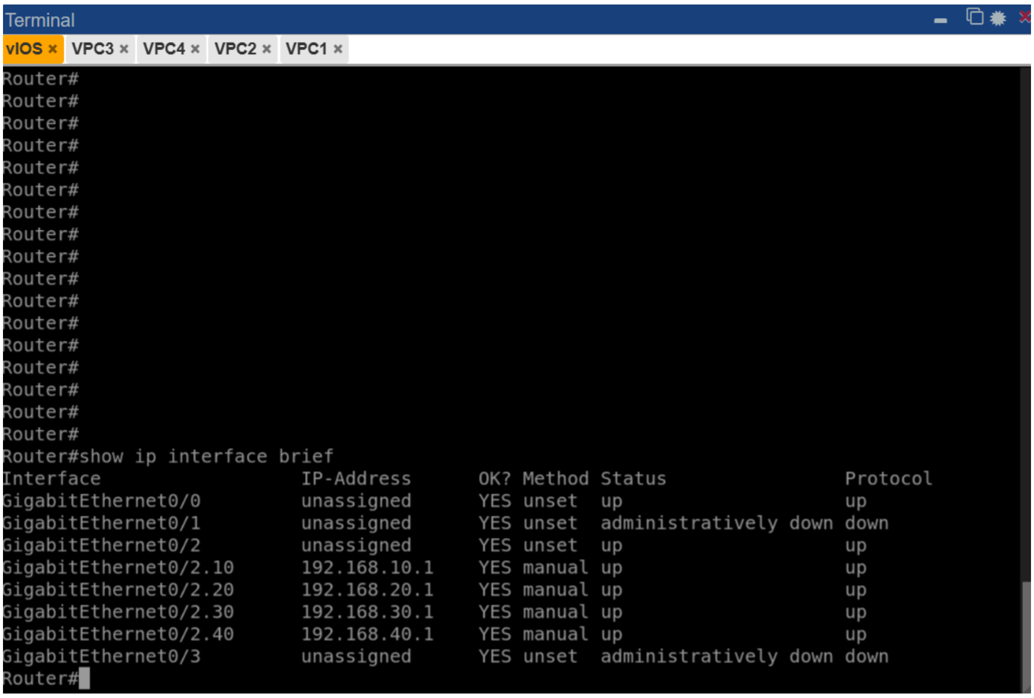


Gambar 16: Konfigurasi IP Static pada vpcs4

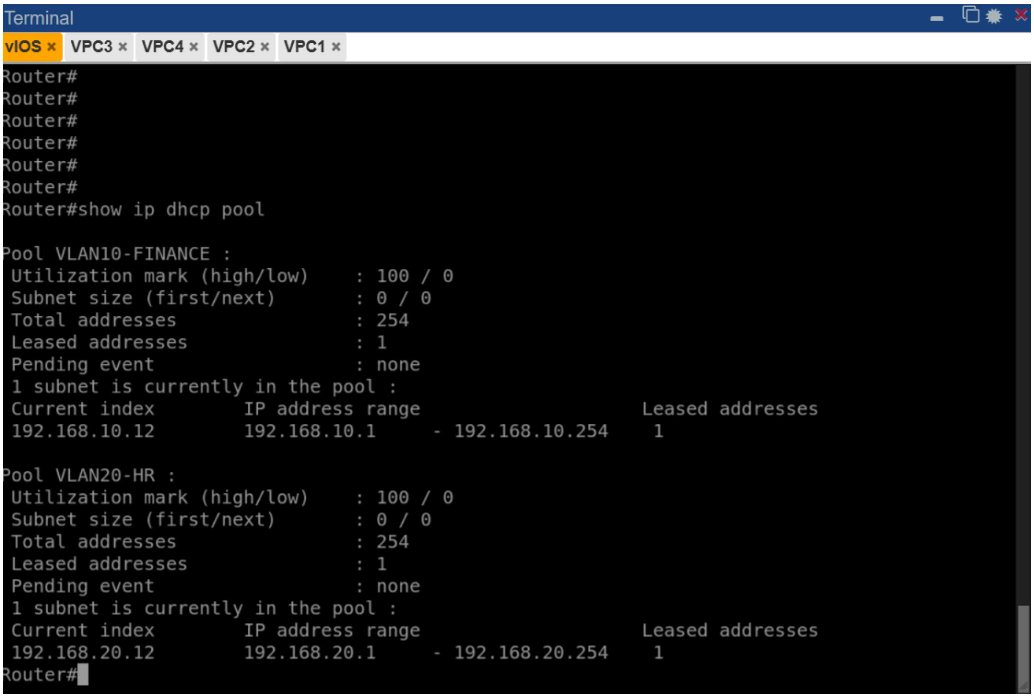


Gambar 17: Konfigurasi IP DHCP pada vpcs1 dan vpcs2

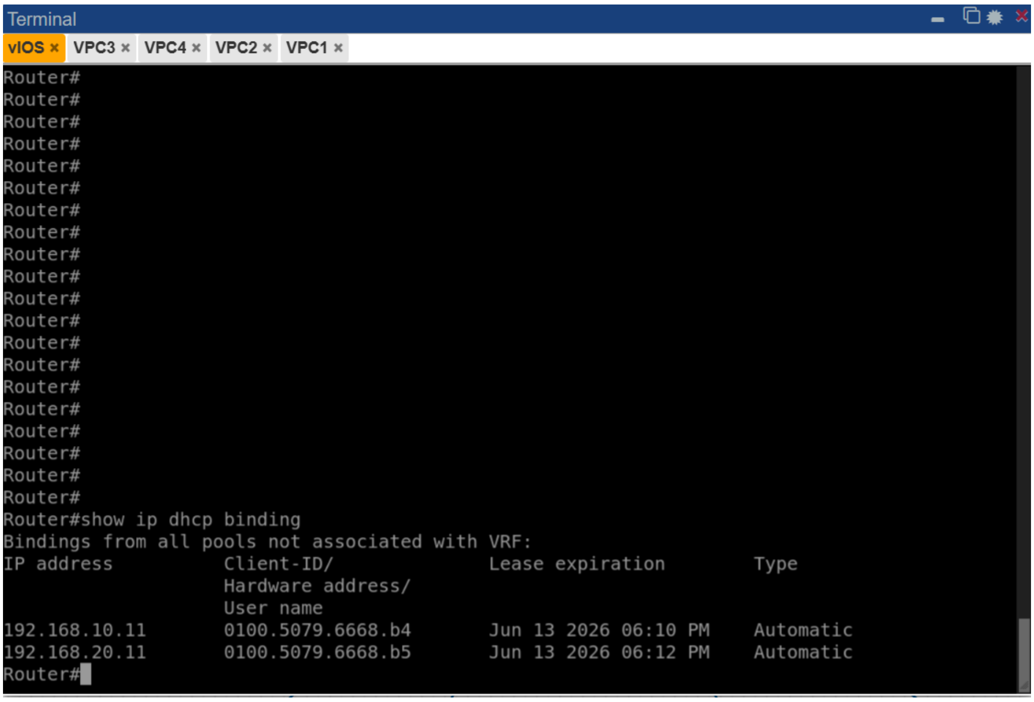
10. Verifikasi Cisco Router



Gambar 18: Verifikasi Router



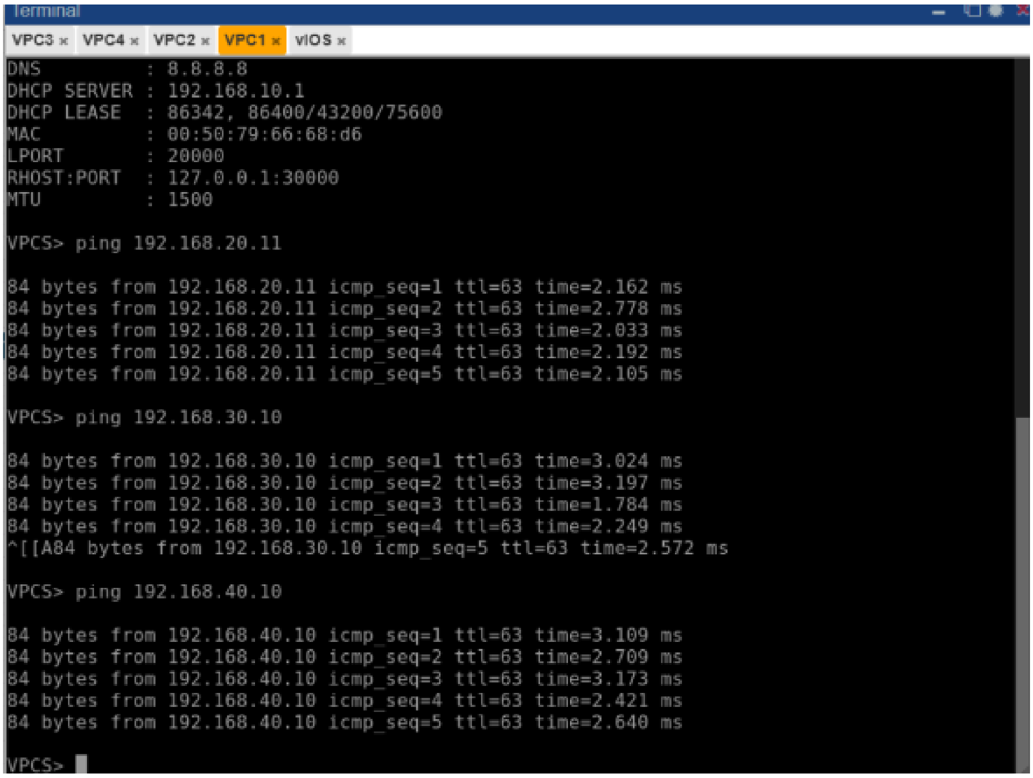
Gambar 19: Cek IP DHCP Pool



```
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.10.11       0100.5079.6668.b4       Jun 13 2026 06:10 PM    Automatic
192.168.20.11       0100.5079.6668.b5       Jun 13 2026 06:12 PM    Automatic
Router#
```

Gambar 20: Cek IP DHCP Binding

11. Pengujian dari VPCS

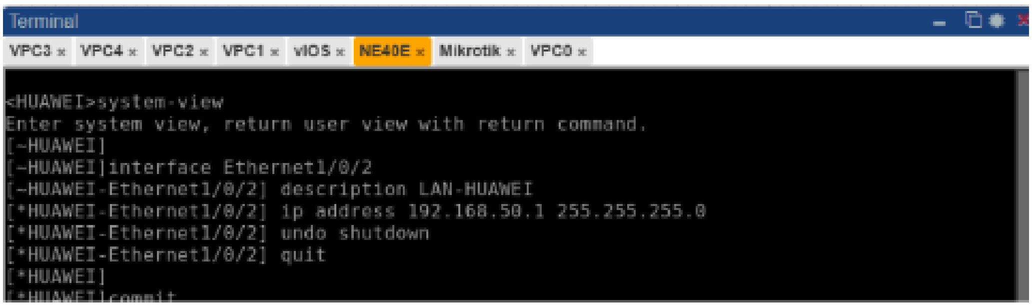


```
DNS         : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86342, 86400/43200/75600
MAC         : 00:50:79:66:68:d6
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=2.162 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.778 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.033 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.192 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.105 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=3.024 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.197 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=1.784 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.249 ms
^[[A84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.572 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.109 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.709 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.173 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.421 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.640 ms

VPCS>
```

Gambar 21: Uji dari vpcs

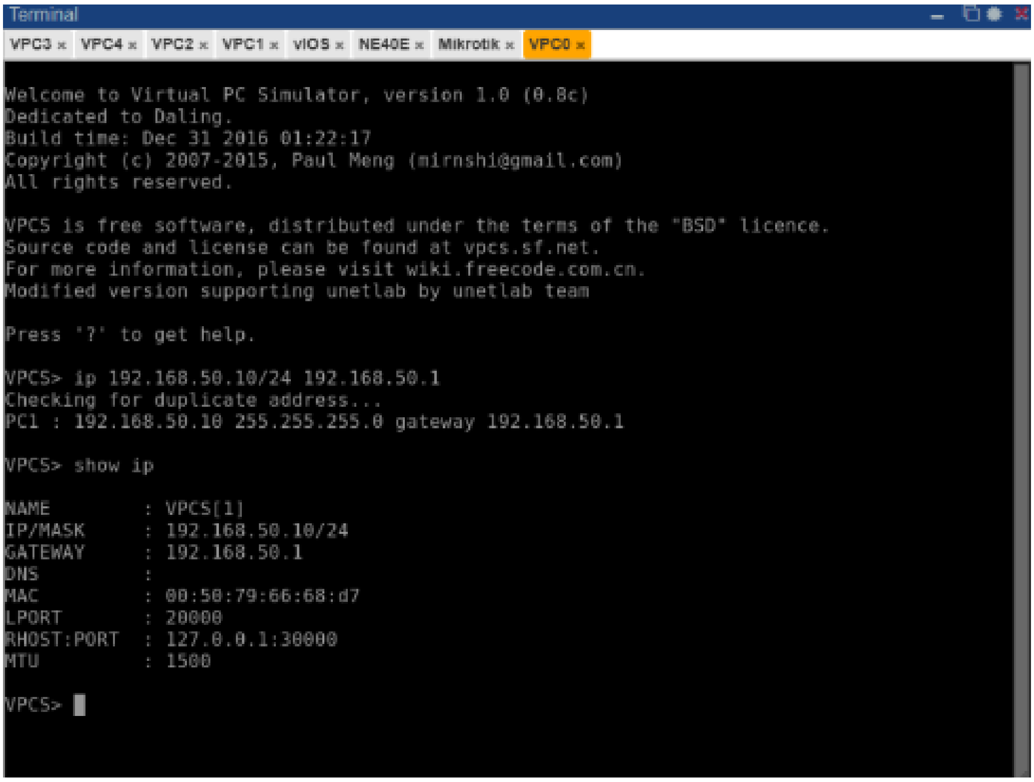
1.3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Konfigurasi Lan Huawei



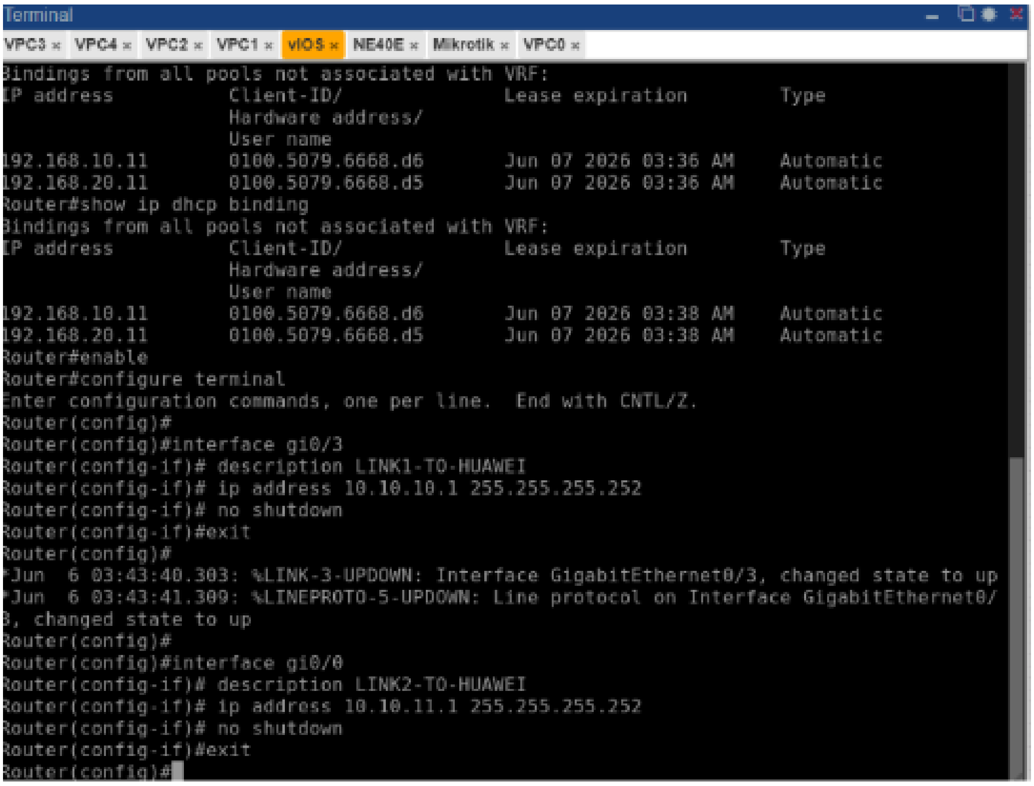
Gambar 22: Konfigurasi Lan Huawei

2. Konfigurasi IP VPC Lan Huawei



Gambar 23: Set IP VPC Lan

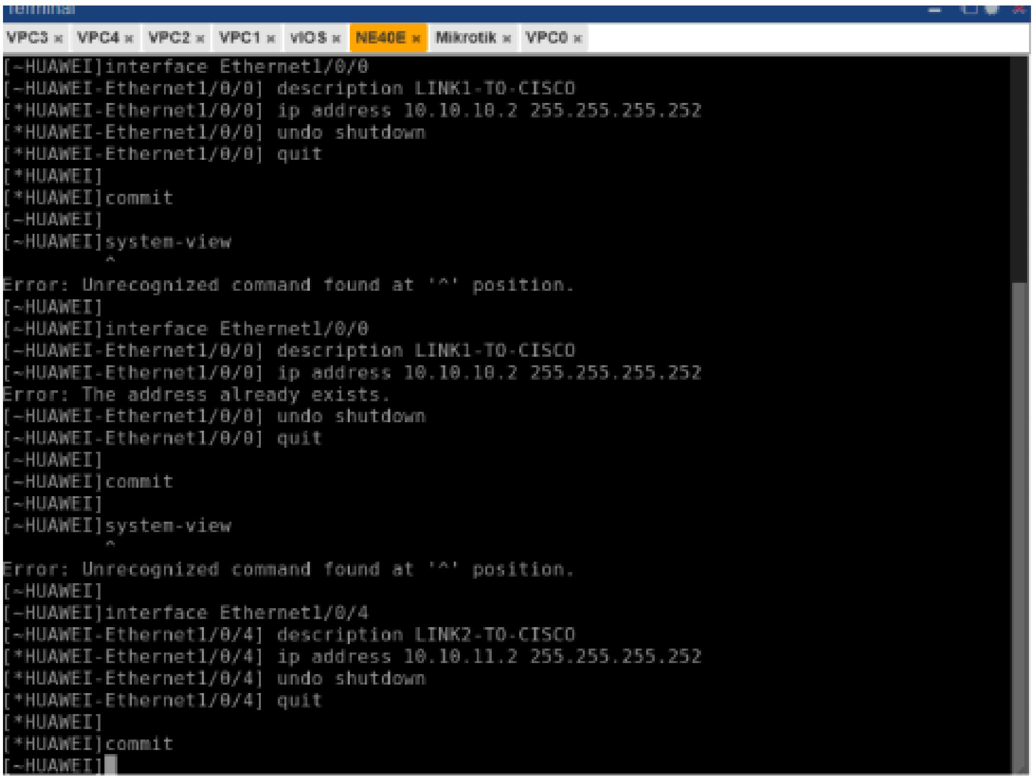
3. Konfigurasi IP link cisco <-> Huawei link 1



```
Terminal
VPC3 × VPC4 × VPC2 × VPC1 × vIOS × NE40E × Mikrotik × VPC0 ×
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.10.11       0100.5079.6668.d6       Jun 07 2026 03:36 AM    Automatic
192.168.20.11       0100.5079.6668.d5       Jun 07 2026 03:36 AM    Automatic
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.10.11       0100.5079.6668.d6       Jun 07 2026 03:38 AM    Automatic
192.168.20.11       0100.5079.6668.d5       Jun 07 2026 03:38 AM    Automatic
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gi0/3
Router(config-if)# description LINK1-TO-HUAWEI
Router(config-if)# ip address 10.10.10.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun  6 03:43:40.303: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun  6 03:43:41.309: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/
3, changed state to up
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# description LINK2-TO-HUAWEI
Router(config-if)# ip address 10.10.11.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
```

Gambar 24: Konfigurasi IP Link cisco ke Huawei

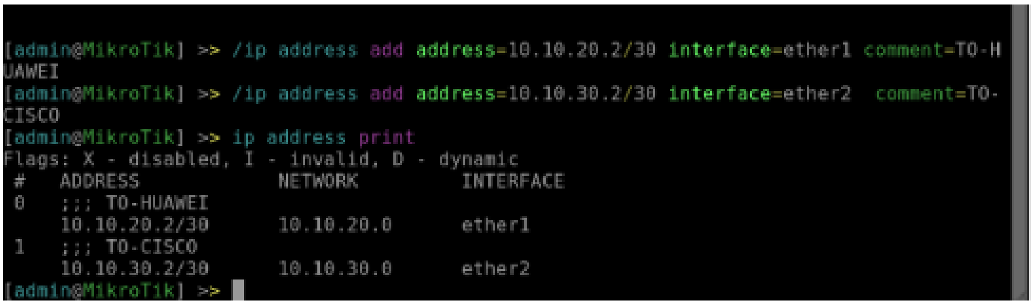
4. Konfigurasi IP Link Cisco <-> Huawei Link 2



```
Terminal
VPC3 × VPC4 × VPC2 × VPC1 × vIOS × NE40E × Mikrotik × VPC0 ×
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0] description LINK1-TO-CISCO
[*HUAWEI-Ethernet1/0/0] ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0] undo shutdown
[*HUAWEI-Ethernet1/0/0] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]
[~HUAWEI]system-view
              ^
Error: Unrecognized command found at '^' position.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0] description LINK1-TO-CISCO
[~HUAWEI-Ethernet1/0/0] ip address 10.10.10.2 255.255.255.252
Error: The address already exists.
[~HUAWEI-Ethernet1/0/0] undo shutdown
[~HUAWEI-Ethernet1/0/0] quit
[~HUAWEI]
[~HUAWEI]commit
[~HUAWEI]
[~HUAWEI]system-view
              ^
Error: Unrecognized command found at '^' position.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4] description LINK2-TO-CISCO
[*HUAWEI-Ethernet1/0/4] ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4] undo shutdown
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 25: Konfigurasi IP Link cisco ke Huawei

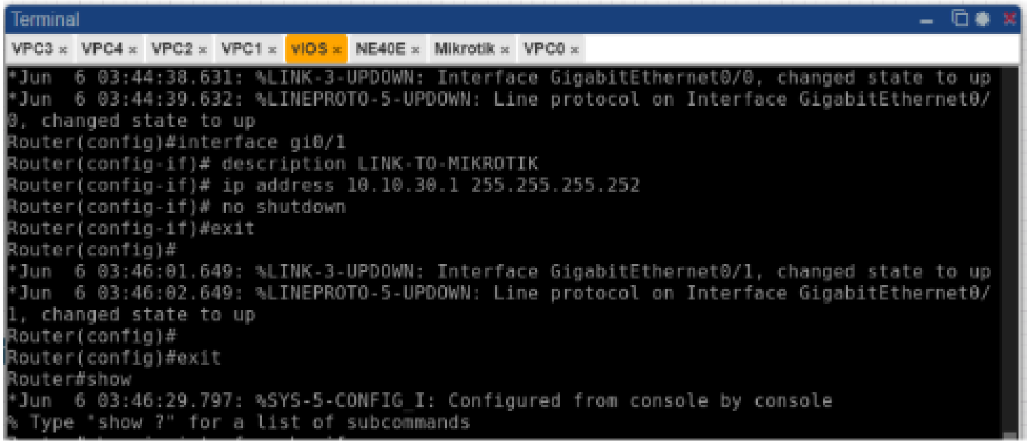
5. Konfigurasi IP link Huawei <-> Mikrotik



```
[admin@MikroTik] >> /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-H
UAWEI
[admin@MikroTik] >> /ip address add address=10.10.30.2/30 interface=ether2  comment=TO-
CISCO
[admin@MikroTik] >> ip address print
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; TO-HUAWEI
     10.10.20.2/30      10.10.20.0      ether1
 1   ;;; TO-CISCO
     10.10.30.2/30      10.10.30.0      ether2
[admin@MikroTik] >>
```

Gambar 26: Konfigurasi IP Link Huawei ke Mikrotik

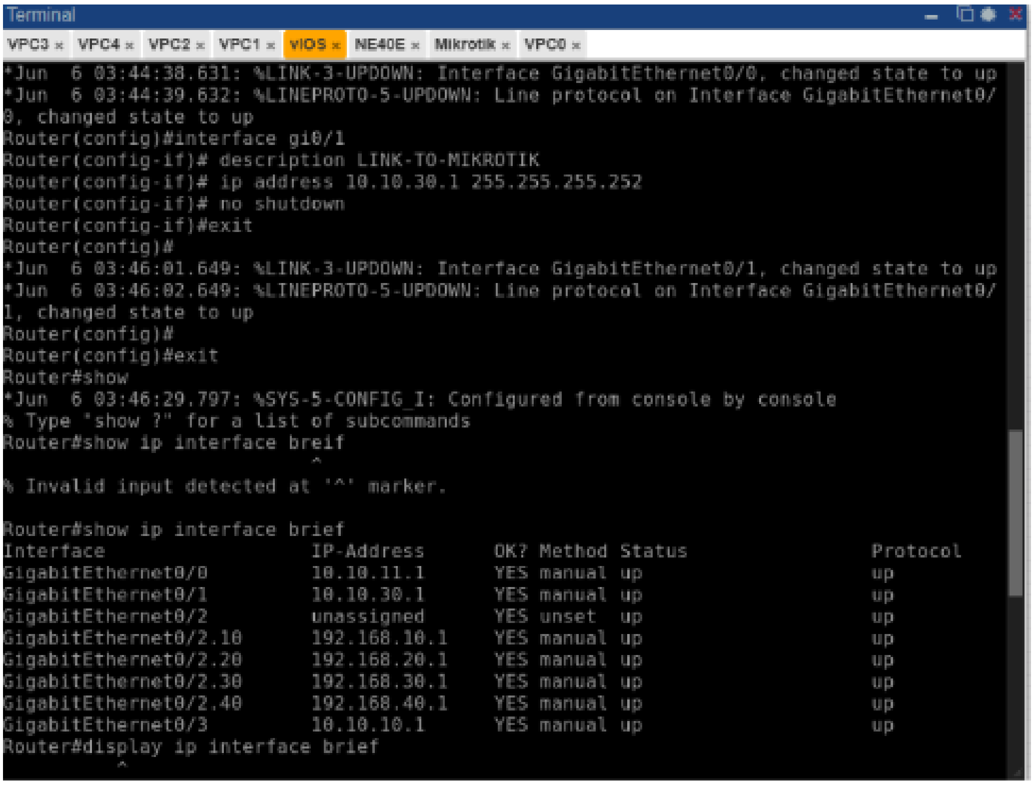
6. Konfigurasi IP link Cisco <-> Mikrotik



```
Terminal
VPC3 × VPC4 × VPC2 × VPC1 × vIOS × NE40E × Mikrotik × VPC0 ×
*Jun  6 03:44:38.631: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun  6 03:44:39.632: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/
0, changed state to up
Router(config)#interface gi0/1
Router(config-if)# description LINK-TO-MIKROTIK
Router(config-if)# ip address 10.10.30.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun  6 03:46:01.649: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun  6 03:46:02.649: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/
1, changed state to up
Router(config)#
Router(config)#exit
Router#show
*Jun  6 03:46:29.797: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
```

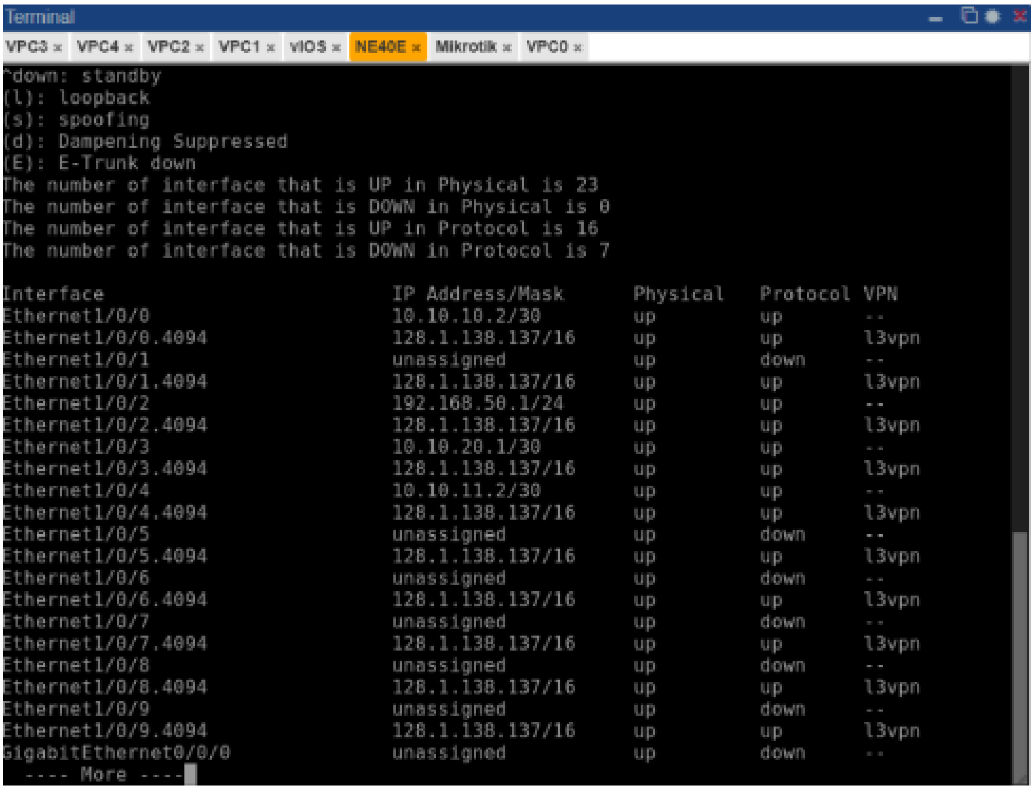
Gambar 27: Konfigurasi IP Link cisco ke Mikrotik

7. Verifikasi Interface



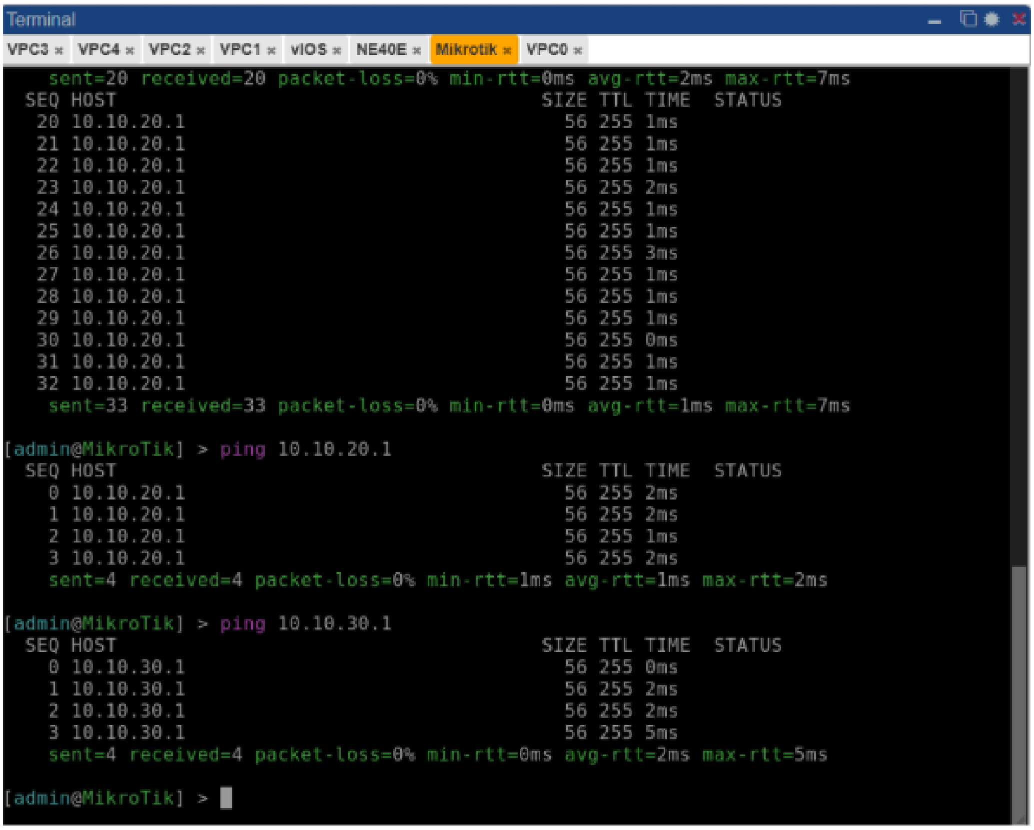
```
Terminal
VPC3 × VPC4 × VPC2 × VPC1 × vIOS × NE40E × Mikrotik × VPC0 ×
*Jun  6 03:44:38.631: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun  6 03:44:39.632: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/
0, changed state to up
Router(config)#interface gi0/1
Router(config-if)# description LINK-TO-MIKROTIK
Router(config-if)# ip address 10.10.30.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun  6 03:46:01.649: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun  6 03:46:02.649: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/
1, changed state to up
Router(config)#
Router(config)#exit
Router#show
*Jun  6 03:46:29.797: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Router#show ip interface breif
                          ^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         10.10.11.1      YES manual up                    up
GigabitEthernet0/1         10.10.30.1      YES manual up                    up
GigabitEthernet0/2         unassigned      YES unset  up                    up
GigabitEthernet0/2.10      192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20      192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30      192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40      192.168.40.1    YES manual up                    up
GigabitEthernet0/3         10.10.10.1      YES manual up                    up
Router#display ip interface brief
         ^
```

Gambar 28: Hasil Verifikasi Interface Cisco Router



Gambar 29: Hasil Verifikasi Interface Huawei Router

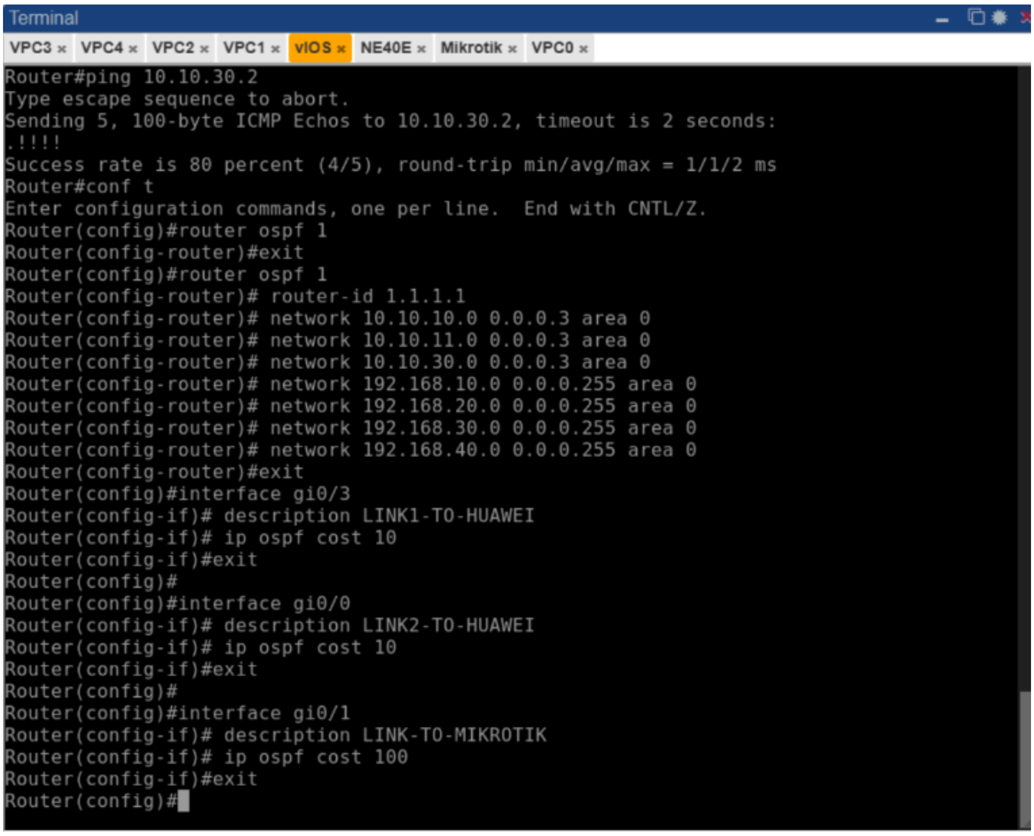
8. Verifikasi ping Antar Router



Gambar 30: Hasil Verifikasi ping antar router

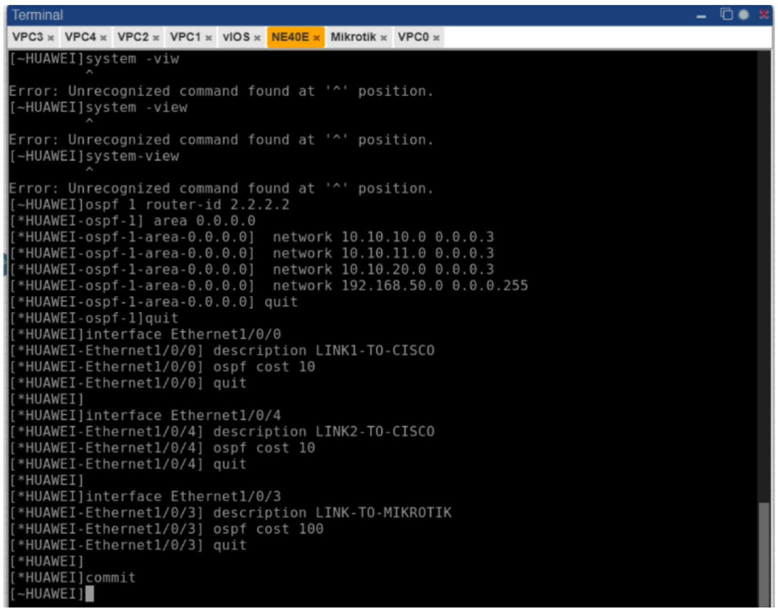
1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konfigurasi ospf pada Cisco Router



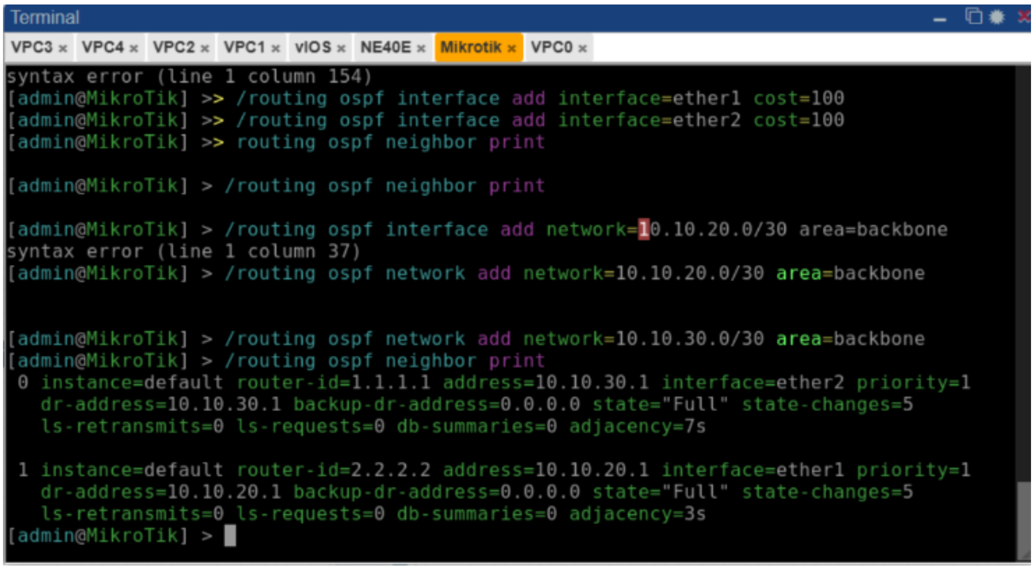
Gambar 31: Konfigurasi ospf pada Router Cisco

2. Konfigurasi OSPF pada Huawei Router



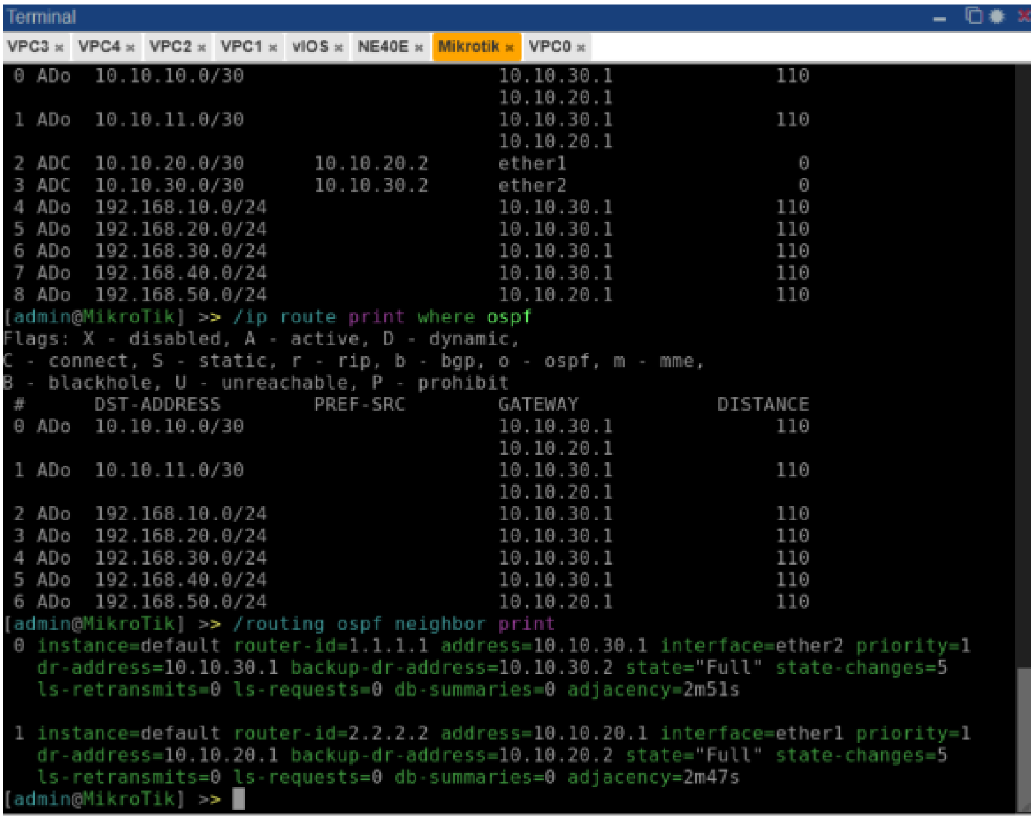
Gambar 32: Konfigurasi ospf pada Router Huawei

3. Konfigurasi OSPF pada MikroTik



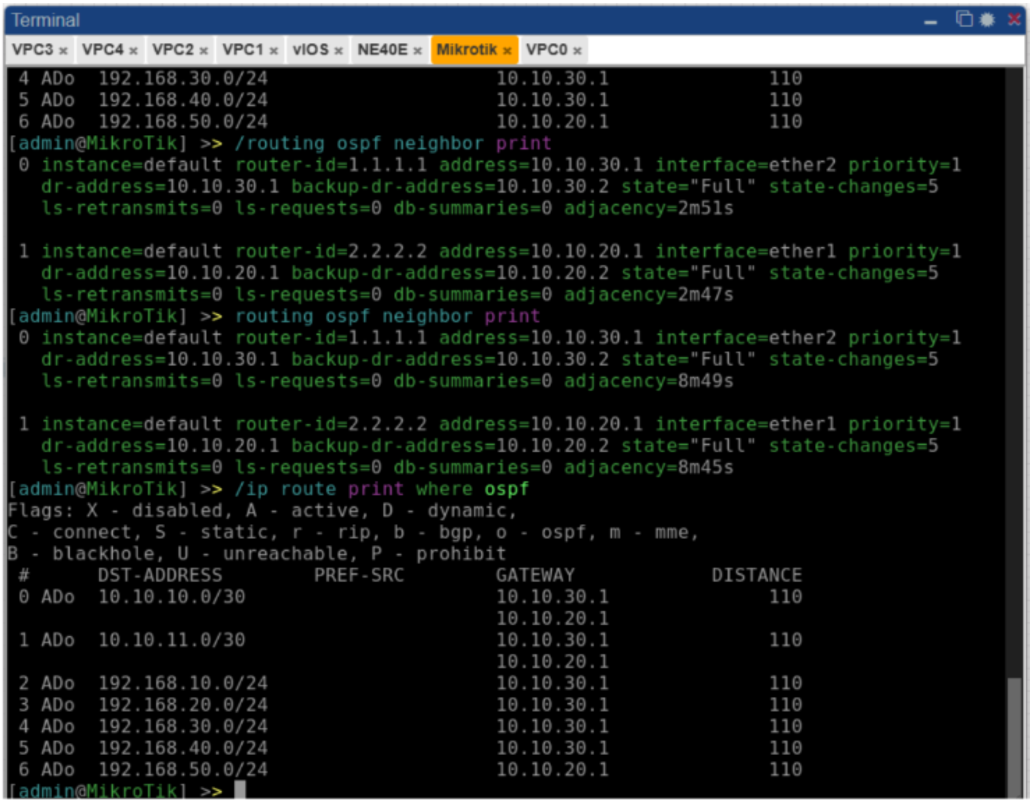
Gambar 33: Konfigurasi ospf pada mikrotik

4. Verifikasi OSPF Neighbor



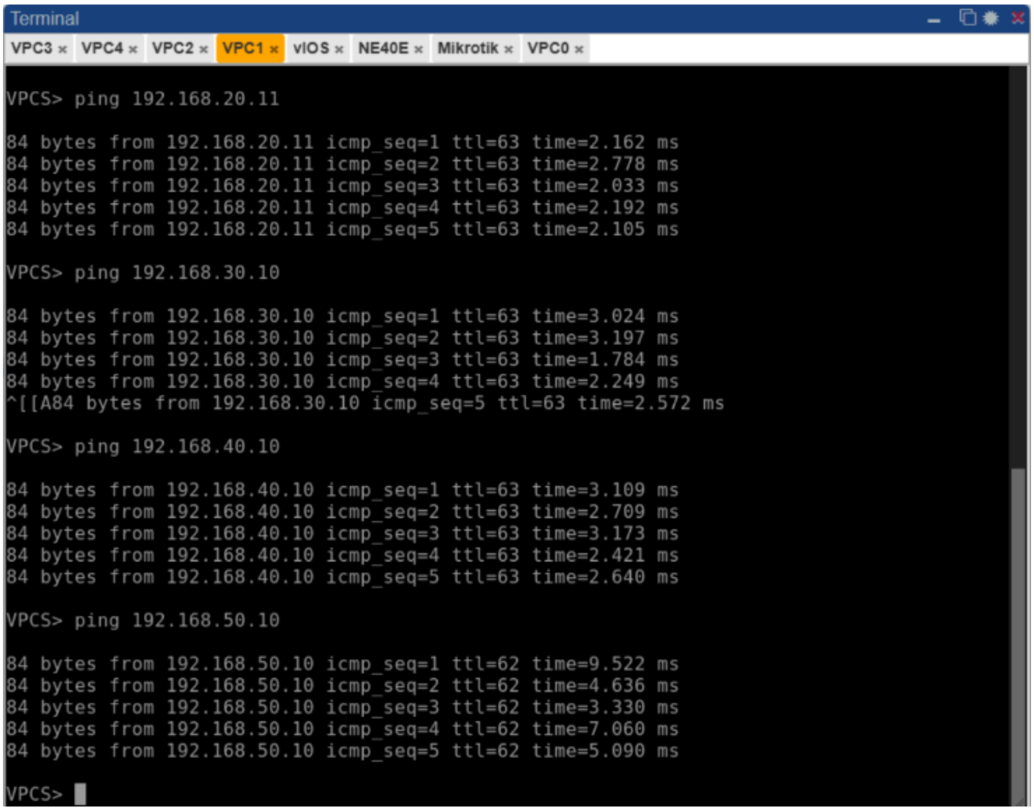
Gambar 34: Verifikasi OSPF milik tetangga

5. Verifikasi Routing Table



Gambar 35: Verifikasi Routing Table

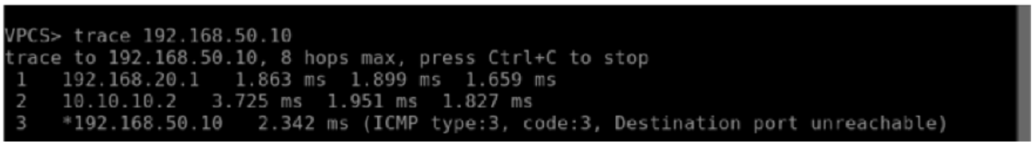
6. Verifikasi Koneksi end-to-end



Gambar 36: Verifikasi koneksi

1.5 Pengujian Failover OSPF

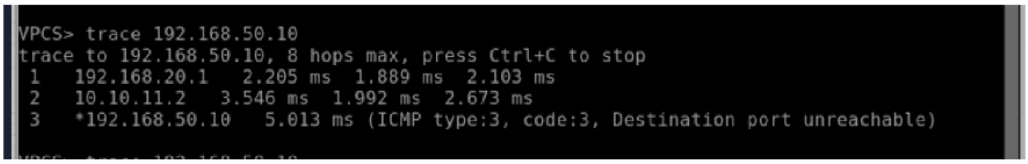
1. Kondisi normal: Dua link cisco-huawei aktif



```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.20.1   1.863 ms  1.899 ms  1.659 ms
 2   10.10.10.2   3.725 ms  1.951 ms  1.827 ms
 3   *192.168.50.10   2.342 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 37: Hasil Show dua link cisco-huawei aktif

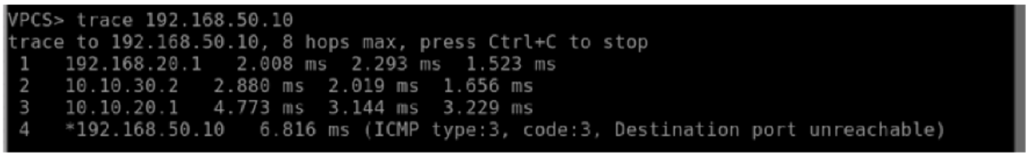
2. Kondisi 1 link cisco-huawei mati



```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.20.1   2.205 ms  1.889 ms  2.103 ms
 2   10.10.11.2   3.546 ms  1.992 ms  2.673 ms
 3   *192.168.50.10   5.013 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 38: Hasil Show satu link cisco-huawei mati

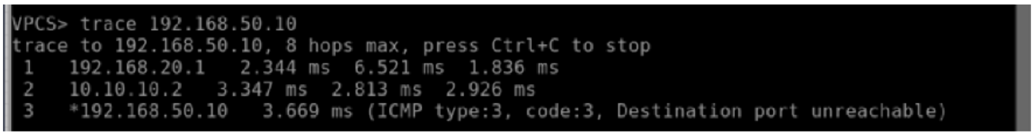
3. Kondisi dua link cisco-huawei mati



```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.20.1   2.008 ms  2.293 ms  1.523 ms
 2   10.10.30.2   2.880 ms  2.019 ms  1.656 ms
 3   10.10.20.1   4.773 ms  3.144 ms  3.229 ms
 4   *192.168.50.10   6.816 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 39: Hasil Show dua link cisco-huawei mati

4. Mengaktifkan kembali dua link cisco huawei



```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.20.1   2.344 ms  6.521 ms  1.836 ms
 2   10.10.10.2   3.347 ms  2.813 ms  2.926 ms
 3   *192.168.50.10   3.669 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 40: Hasil Show pengaktifan dua link cisco-huawei

2 Analisis Hasil Percobaan

Pada praktikum ini seluruh konfigurasi berhasil dilakukan sesuai tujuan yang ditetapkan. Cisco Switch berhasil dikonfigurasi dengan VLAN 10, 20, 30, dan 40 serta trunk port yang mampu membawa seluruh VLAN menuju Cisco Router. Implementasi Router-on-a-Stick pada Cisco Router memungkinkan komunikasi antar-VLAN melalui subinterface yang berfungsi sebagai gateway masing-masing jaringan. Selain itu, DHCP Server berhasil memberikan alamat IP secara otomatis kepada host pada VLAN 10 dan VLAN 20, sedangkan VLAN 30 dan VLAN 40 menggunakan alamat IP statis. Pengujian konektivitas menunjukkan bahwa seluruh host dapat berkomunikasi dengan gateway maupun perangkat pada VLAN lain.

Konfigurasi alamat IP antarrouter pada Cisco Router, Huawei Router, dan MikroTik Router juga berhasil diterapkan. Seluruh router dapat saling melakukan *ping* pada jaringan yang terhubung langsung, menunjukkan bahwa konfigurasi addressing dan konektivitas fisik telah berjalan dengan baik.

Jaringan LAN Huawei pada subnet 192.168.50.0/24 juga berhasil terhubung dengan perangkat lain dalam topologi.

Implementasi OSPF pada ketiga router berhasil membentuk hubungan *neighbor* sehingga setiap router dapat mempelajari rute jaringan lain secara dinamis. Routing table pada Cisco, Huawei, dan MikroTik menunjukkan bahwa seluruh network berhasil dipertukarkan melalui OSPF. Pengujian konektivitas end-to-end membuktikan bahwa seluruh VLAN pada sisi Cisco dapat berkomunikasi dengan LAN Huawei, demikian pula sebaliknya.

Pada pengujian failover, OSPF mampu melakukan konvergensi secara otomatis ketika salah satu maupun kedua link utama Cisco–Huawei dinonaktifkan. Saat kedua link utama gagal, trafik berpindah melalui jalur cadangan yang melewati MikroTik. Setelah link utama diaktifkan kembali, OSPF mengembalikan jalur routing ke rute dengan cost yang lebih rendah sehingga komunikasi tetap berlangsung tanpa perlu konfigurasi ulang secara manual.

3 Hasil Tugas Modul

Hasil Tugas Modul terdapat pada Github kelompok yang dapat diakses melalui link yang tertera.

4 Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, dapat disimpulkan bahwa konfigurasi VLAN, trunking, Router-on-a-Stick, DHCP Server, addressing antarrouter, serta routing dinamis OSPF pada lingkungan multivendor berhasil diterapkan dengan baik. Seluruh perangkat mampu saling berkomunikasi sesuai topologi yang dirancang dan seluruh jaringan berhasil dipertukarkan melalui OSPF. Selain itu, mekanisme failover OSPF terbukti mampu menyediakan redundansi jaringan dengan mengalihkan jalur komunikasi secara otomatis ke link cadangan ketika link utama mengalami gangguan. Dengan demikian, tujuan praktikum untuk memahami implementasi VLAN, routing antarjaringan, OSPF multivendor, dan failover routing telah tercapai dengan baik.

5 Lampiran

5.1 Dokumentasi saat praktikum

Gambar 41